

Representation Convergence - Project Status

Thesis: independently trained models share representations up to a linear transformation, and that convergence can be deployed. All models frozen; only closed-form linear maps are fitted, except where marked (F).

Models used (all pretrained, frozen unless marked)

Model	Modality / training	Role
GPT-2 (124M)	text, next-token prediction	Exp A subject; E1.1 raw-LM text arm
Pythia-160M	text, next-token (different data/arch lineage)	Exp A counterpart
MobileCLIP-S1	vision, image-text contrastive (phone tier)	Exp B source; F student input
SigLIP 2 base	vision-text contrastive (server tier)	Exp B target; retrieval ceiling; F teacher
DINOv2-base	vision, self-supervised, no language supervision	E-series image arm
bge-m3	text retrieval encoder, contrastive on 1.2B pairs	E1/E2 text arm
MLP heads (trained)	-	B2-bis and F2 nonlinearity controls; F3 unfreezes MobileCLIP last stage (scaffolded)

Experiments and results

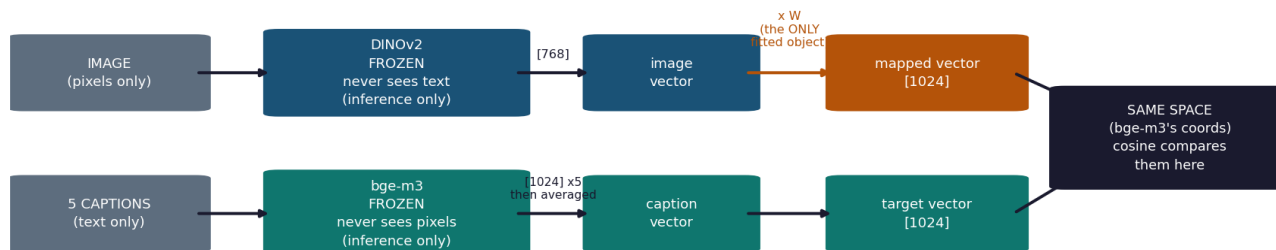
Exp	Question	Setup	Key result	Status
A	Do two LLMs converge?	GPT-2 vs Pythia, 10k WikiText passages, 13 layers, CKA + layer-wise ridge stitching + seeded random-weights control. Data: 7,500 train rows / 768 dims = 9.8	Stitching R2 0.737-0.798 vs random 0.17; CKA 5.3x control; trained and random controls diverge with depth (opposite slopes)	DONE
A2/A3	Is the correspondence structure real?	argmax matrix, lift correction, symmetry, length probes, effective dimensionality	Correspondence elevated but hub-dominated (97% rank-1); lift: 77% within 2 layers; L0 gap = architecture (abs-pos vs RoPE); asymmetry = variance-weighting artifact; eff.dim GPT-2 2.2-10.9 vs Pythia 9.3-92.7 of 768	DONE
B	Are two vision encoders linearly interchangeable?	MobileCLIP -> SigLIP, 4k COCO val2017 pairs, ridge vs Procrustes vs MLP. Data: 3,000 train / 512 = 5.9 (bottom of the 5-10 band)	Ridge R2 0.592 / cosine 0.900; Procrustes fails (0.038) -> anisotropy, confirmed by W's 69x singular-value spread; MLP +0.003 -> relationship is linear	DONE
B3/D	Is it usable?	Retrieval + membership on shared Qdrant index, fp16 export	R@5 = 95.1% of ceiling through a 0.79 MB adapter; raw = chance; cross-tier membership R@1 0.987, gap +0.075 (98.7% positive); fp16 bit-exact	DONE, deployed
B6-B10	Quantization tiers	int8 / int4 adapter + embedding quantization	int8 lossless for this use; documented tier table	DONE
E1	Cross-modal: vision-only vs text-only (PRH's actual claim)	DINOv2 -> bge-m3, 8k COCO train2017, averaged captions, direction- and protocol-matched SigLIP ceiling, pre-registered bands. Data: 7,000 train / 768 = 9.1	R2 0.511, shuffle gap 0.641; R@1 0.358 = 40.9% of matched ceiling (0.875); verdict MEASURABLE BUT WEAK, robust across caption protocols (24.7% / 40.9% both in band); crossing modalities costs ~half of R@1 vs same-modality	DONE
E1 sweep	Does cross-modal alignment grow with encoder capacity? (PRH's actual claim)	Same protocol, DINOv2-large with cls+patch pooling (CLS token concatenated with the patch mean, width 2048). Data: 22,755 train / 2,048 = 11.1 rows/dim; eval 1,000	R2 0.609 - above the same-modality vision pair's 0.592 - cosine 0.912, R@1 0.504 = 58.3% of a matched ceiling (0.865), shuffle gap 0.709. Confounded: encoder size AND pooling both changed, so this is not yet a capacity measurement; small and base at cls+patch are the matched points still to run	PARTIAL
E1.1	Does the text side need similarity training?	Matched pair: replace bge-m3 with GPT-2 mean-pooled; all else identical (8k pairs, 7,000 / 768 = 9.1); target geometry measured before fitting	Predicted split observed: R2 retains 74% (0.377), R@1 retains 27% (0.095, still 95x chance); GPT-2 eff.rank 6.5/768, pair-cos +0.999 -> cosine degenerate (reads 1.000); ridge absorbs anisotropy, cosine cannot	DONE
E1.2	Can label-free geometry rescue GPT-2's space?	6 arms: raw / centered / all-but-top-k / whitened, transforms fitted on train rows only; validated on synthetic control (raw 0.375 -> centered 1.000)	Stake: separates 'correspondence absent' from 'isotropy absent'; caution for cosine-based alignment metrics in the PRH literature	BUILT, not yet run
E2	Interactive demo	image -> nearest captions; text -> nearest images; one W, both directions; negative control included	Working demo over 500 unseen val2017 images	BUILT

E3	Functional stitching: can a consumer USE the map?	SigLIP text tower consumes translated MobileCLIP vectors; 3 arms + data-efficiency curve; internal-splice scaffold	Stitching penalty at R@5 = 0.049 -> near-transparent; 'preserved vs usable' separated	DONE (splice sc afforded)
F1-F2	Was the MLP control data-starved?	38k-image teacher corpus, no captions (SigLIP 2, checkpoint teacher-tagged); 38,000 / 512 = 74 rows/dim, 0.029 examples/param for the MLP; relational loss	MLP converges to EXACTLY the linear solution (0.9052 = 0.9052, gain -0.0000, plateau by epoch 10) -> bottleneck confirmed twice; data-starvation reading refuted	DONE
F5	Is the refit linear adapter better than the shipped one?	Same ridge refit on 38k rows (74/dim); eval on 5,000 fresh queries, 5 disjoint 1k galleries unseen by both maps; paired bootstrap	All gains REAL: R@1 +0.024 [+0.016,+0.033], R@5 +0.012, R@10 +0.011; same 0.79 MB artifact; 90%-of-ceiling gate approached (89.0-90.6%, dataset-dependent), not cleanly met	DONE
F3-F4	Can the ceiling itself move?	Unfreeze MobileCLIP last stage + head; compare-and-decide with bootstrap ship gate	Gated on F2's verdict (now satisfied); deliberately unrun - trades the adapter's seconds-refit property for weights	BUILT, gated

How the E-series flow works (and what is NOT trained)

A layer outputs one vector per token, not one vector. A transformer stage reads a set of token vectors and writes a set of the same shape, so a five-token passage through GPT-2 yields 13 layers x 5 tokens = 65 vectors. Masked mean pooling collapses the TOKEN axis only, separately within each layer - real tokens are summed and divided by their own count, pads excluded from both - leaving 13 vectors per passage, one per layer. The layer axis survives because depth is what Experiment A measures. Worked example on 4 coordinates with 3 real tokens and one pad: sum [6, 9, 6, 6] divided by 3 gives [2, 3, 2, 2]; including the pad would have given [3.75, 4.5, 3.75, 3.75]. Pooling is unavoidable because two tokenizers do not agree on what the tokens are - token-to-token correspondence does not exist between GPT-2 and Pythia, passage-to-passage does. **Neither model ever sees the other modality.** Images enter DINOv2 only; captions enter bge-m3 only. The 'image-caption pair' lives in the row index of two arrays - row i of the image array and row i of the caption array describe the same COCO image - which is bookkeeping for the fit, not an input to either network. **Direction of the map.** W converts DINOv2 vectors [768] into bge-m3's coordinate system [1024] because the space you land in must be the space you search in: image vectors and caption vectors then live in one index and cosine ranks across the modality boundary. One W serves both search directions - for text-to-image, the gallery images are pushed through W once and a typed query meets them natively. Fitting from the narrower side (768 in) also needs fewer rows for the same statistical power. **No training occurred.** Both models are downloaded frozen and run under no_grad; W is a closed-form ridge solution, not gradient descent. After every A/B/E experiment the model weights are bit-identical to the downloads - delete W and nothing remains. The experiments MEASURED a pre-existing relationship; they did not train one in. (The F series is the deliberate exception - the contrast condition, not a lapse: F2 trains a separate MLP head beside the frozen encoders, and F3 - built but deliberately unrun - is the only notebook that would modify a pretrained model's own weights. As of this report every pretrained model in the project remains bit-identical to its download.)

E1 data flow - the two models never see the other modality; only their OUTPUTS meet



the "pairing" (image i <-> its captions) lives in the ROW INDEX of two arrays - it is bookkeeping, not an input to either model

The same flow as arithmetic (toy sizes: image dim 3, text dim 2)

```

image vec a = [0.9, 0.1, 0.4]      (DINOv2 output, row i)
target t = [0.60, 0.80]           (bge-m3 caption avg, row i)

[ 0.5  0.6]
W (3x2)= [ 0.4 -0.2]  fitted by ridge on 7,000 such rows
[ 0.3  0.9]

predicted u = a x W
= [0.9*0.5+0.1*0.4+0.4*0.3,
   0.9*0.6+0.1*-0.2+0.4*0.9]
= [0.61, 0.88]

cos(u, t) = (0.61*0.60+0.88*0.80)
/ (|u| |t|) = 0.999 -> row i found
  
```

What exists before and after the experiment

DINOv2 weights	unchanged - bit-identical
bge-m3 weights	unchanged - bit-identical
W (768 x 1024 matrix)	NEW - the only thing created; closed-form ridge, no gradient descent, seconds to fit

delete W and you are back to exactly the two downloaded models - the experiment MEASURED a relationship, it did not train one in

Figure: the E1 flow, the same flow as worked arithmetic (toy dimensions), and the before/after inventory - two unchanged models plus one new matrix.

Caption protocol in the E-series

COCO supplies **five independent human captions per image**, written by five annotators; five is the dataset's structure, not a tuned choice. The decision was whether to use one or all. A single caption carries that annotator's emphasis - one names the activity, another the setting - so the target vector mixes image content with individual phrasing. Averaging the five embeddings cancels the idiosyncrasies and keeps what they agree on, making the target 'what the image is about' rather than 'what one person wrote'. The effect is large: held-out R2 rises 0.278 to 0.511 and R@1 0.160 to 0.358. The same averaging must be applied to the ceiling, because it makes the task easier for any method - SigLIP itself scores 0.648 on single captions and 0.875 on averaged ones. Scoring an averaged system against a single-caption ceiling produced an inflated 55.2 percent; matched properly it is 40.9. Both protocols are reported, and both fall in the same pre-registered band (24.7 and 40.9 percent), so the verdict does not depend on the choice; note that the standard COCO benchmark convention is one caption per query, so averaging is a deliberate, disclosed deviation.

A note on alpha

ALPHA_GRID in the E-series is a **search over regularization strength**, not a learning-rate schedule: ridge has no training loop, no gradients and no steps - W is one closed-form solve. Each candidate is scored on a validation slice carved from the training rows only, the winner is fixed, and one final fit follows. How much alpha matters is itself diagnostic: in the dinov2-large run every candidate scored 0.596 (22,755 rows against 2,048 dimensions is well-conditioned), while in E1.1 alpha = 10 gave R-squared -0.741 because GPT-2's near-degenerate space is severely ill-conditioned. Genuine learning rates appear only in the F series (F2 at 1e-3 with cosine annealing, F3 at 1e-4).

The scale sweep: alignment vs encoder capacity

The cross-modal protocol was repeated across DINOv2 small, base and large - 22M to 304M parameters, a 13.8-fold range - holding the text encoder, data, pooling and ridge protocol fixed, and scaling the number of pairs with output width so that rows per input dimension is 11.1 at every point. Share of a matched, per-run ceiling at R@1: **40.9, 53.5 and 58.3 percent**. Every measure rises monotonically - R-squared (0.516, 0.580, 0.609), cosine, all three recall levels, and the shuffle-control gap (0.616, 0.682, 0.709). Against log parameters, R@1 gains 15.3 points of ceiling per decade (fit R-squared 0.949). Both base and large exceed the same-modality reference of 0.592, on a task that crosses modalities without joint training. The returns are compressive - 4.0x params buys 12.6 points, the next 3.5x buys 4.8 - so the fit summarises three points rather than stating a law, and nothing is claimed beyond 304M. This discharges the limitation that the study could sample the hypothesis but not observe its slope.

Isotropy, not similarity training (E1.2)

Section C.9 concluded that usable retrieval geometry requires similarity training. E1.2 refutes that. Four label-free geometric corrections were applied to GPT-2's caption space - centering, top-component removal at k = 1, 3 and 10, and whitening - each estimated on training targets only. Whitening lifts R@1 from 0.130 to **0.478**, reaching **102.6 percent** of an encoder contrastively trained on 1.2 billion text pairs. The correspondence was present in the raw language-model representations all along; only the coordinate frame concealed it. Isotropy behaves as a continuous dial rather than a threshold - effective rank against R@1 has rank correlation +0.94 across the six arms - and centering alone fails instructively, driving mean pair-cosine from +0.999 to -0.019 while leaving effective rank at 6.5 and retrieval unimproved. The whitened arm records held-out R-squared of 0.014 alongside the best retrieval measured, completing the project's argument that a similarity metric is interpretable only relative to the geometry of the space it is computed in.

Unsupervised recovery (C1.1)

The final piece of supervision - the pairing itself - was removed. Two sets of 1,500 vectors from the same images, one side shuffled and the true permutation sealed until scoring, matched by entropic Gromov-Wasserstein using only each space's own internal distance matrix. Exact matching reaches **39.7 percent against a chance rate of 0.067** - 595x - and the map fitted from that assignment reaches R@1 0.252, 40 percent of the jointly-trained ceiling, with verification AUC 0.906. The correspondence between two independently trained encoders is therefore partially self-identifying: discoverable from geometry, without being told. Two negatives are reported with it. Iterative refinement degraded the solution monotonically, converging stably on its own errors. And whitening - the transform that rescues cosine retrieval in E1.2 - collapses matching to chance, because Gromov-Wasserstein matches structure to structure and uneven variance is what makes a point distinctive relative to its neighbours. Isotropy helps when reading a correspondence and hurts when finding one: the same property, opposite prescriptions, both measured.

The shared hub (G1): the thesis made constructive

Five cached spaces over the same 9,533 images - DINOv2 small, base and large, bge-m3, and GPT-2 - were each given one linear map into a single 512-dimensional whitened-PCA hub. A caption-retrieval head trained on ONE encoder's hub vectors reaches 95.9 and 92.9 percent of natively-fitted heads on the two encoders it never saw, with absolute R@1 rising (0.398, 0.442, 0.468) and a random-map control at chance. One coordinate system serves all five encoders, and a component built in it transfers between encoders that never met - the constructive form of the thesis. Cross-lineage holds (SigLIP 2, 94.2%); cross-ARCHITECTURE holds too and more strongly than expected - ConvNeXt-base reaches 96.7 percent of native, ABOVE the within-family band. The transfer cliff between 512 and 768 hub dimensions is now EXPLAINED, and it is a numerical artifact rather than a finding about representations: the head's coordinates have algebraic rank equal to the source encoder's dimension, so at that width least squares divides by a near-null singular value (0.0054 against 10.79 one step earlier). Seven accounts were falsified first; the eighth predicted the location out of sample and resolved to a single direction on two encoders. Truncating the fit or raising the HEAD's alpha repairs it - so C.13's 'alpha cannot substitute for width', measured on the entry-map alpha, needs narrowing, and the portability/losslessness reading of the cliff does not survive.

The adapter as a decider: three problems, one hierarchy (D.2)

The same 0.79 MB adapter was measured as three different decision rules on 1,000 held-out images. Verification - are these two vectors the same photo - reaches AUC **0.999**. Per-query ranking reaches R@1 **0.987**. Membership under one global threshold - is this phone photo indexed at all - reaches AUC **0.889**, with best F1 0.840. Difficulty rises as the decision becomes more global: a verification negative is a typical impostor, while a membership negative is the best of 500, pushed into the positives by extreme-value statistics. ROC applies to these thresholded tasks and NOT to R@K retrieval; a caption-verification AUC of 0.997 with positive cosine 0.136 against negative -0.024 shows AUC measures separation, not magnitude - the metric-geometry lesson once more.

The encoder ladder (E1.3): C.11 tested as a prediction

Two encoders C.11 had never seen - BERT (masked LM) and Sentence-BERT all-mpnet (contrastive) - were run under the identical protocol, with predictions registered in the notebook before the numbers existed. Prediction 1 confirmed exactly: raw R@1 orders by the target's effective rank with rank correlation **+1.00** (eff.rank 6.5 / 65.0 / 93.4 maps to R@1 0.160 / 0.362 / 0.510), and BERT lands precisely between GPT-2 and the contrastive encoders as registered. Prediction 2 - that whitening collapses the ladder - held only partially, and the miss is the sharper finding: whitening lifts every arm (0.478 / 0.521 / 0.588) but a 0.110 spread remains in the raw ordering. **Isotropy is necessary but not sufficient**: whitening equalises how many directions carry variance but cannot install which directions carry meaning. Contrastive training contributes a coordinate frame - supplied for free by a label-free linear map, and nearly the whole story for the most degenerate space - plus a semantic arrangement that survives whitening and measures the true signature of the objective.

The hub at seven encoders (G1, extended)

Adding the two ladder encoders of E1.3 brings the hub to seven spaces across **four text training objectives** and two modalities. Transfer is UNCHANGED, and that is the useful result: R@1 **0.451** for a head trained on one encoder and applied to an unseen one (verification AUC 0.986), against 0.445 for the five-space hub. The difference is six queries in a thousand - 0.27 standard errors, 95 percent interval -0.038 to +0.050 - so it is not distinguishable from sampling noise. The hub absorbs two further training objectives at no measurable cost to the components already using it. Composition through the hub retains **95.6 percent** of directly-fitted pairwise maps across 36 of 41 ordered pairs. The five exceptions are all the same target: writing INTO GPT-2's collapsed space, at 24.8 percent. Reading OUT of it works at 95.1 percent, and it reaches the bge readout space at 93.9, so every encoder is usable for the deployed direction. The cause was tested rather than assumed: the hub reconstructs the collapsed space perfectly, and the failure comes from ranking within a space whose own items sit at +0.999 cosine from each other, where one extra fitting stage costs more than the gaps are wide. Measured also: shape agreement BEFORE any map exists - cross-modal pairs at rho +0.29, and bge-SBERT highest of all 21 pairs at +0.76 despite sharing no lineage, only an objective.

What has been established

1. Convergence is real and linear. Two LLMs and two vision encoders, trained independently, are related by a single linear map (held-out R2 0.74 / 0.59); an MLP control adds nothing at 3k samples (+0.003) and converges to exactly the linear solution at 38k (-0.0000). **2. It is not rigid.** Rotation-only alignment fails (0.038); the deviation is anisotropy, measured three independent ways (69x singular-value spread; 58x effective-rank gap; Procrustes-cosine 0.775 with R2 0.038). **3. It is usable.** One 0.79 MB matrix gives two encoders one index at 95% of ceiling, plus an unplanned capability (cross-tier membership 0.987) predicted from the theory and then confirmed. **4. It extends across modalities, weakly.** A vision model with no language supervision maps onto a text encoder at 358x chance - 41% of a jointly trained ceiling - with the verdict robust to protocol choice. **5. Metric validity depends on the space's geometry.** In SigLIP's space cosine was the honest metric and R2 understated; in GPT-2's near-1D space cosine reads 1.000

while retrieval runs at 9.5%. No similarity number is interpretable before the anisotropy battery is run - a caution that applies to the PRH literature's own metrics.

Methodological failures caught and converted

Run-1 sample starvation (5.9 rows/dim) -> the ≥ 5 rule and the power gate; synthetic hot-diagonal figure -> replaced with the real hub-dominated matrix and an honesty note; mismatched E1 ceiling (borrowed direction and protocol) -> measured matched ceiling, verdict corrected 55.2% -> 40.9%; wrong teacher checkpoint in F1 (SigLIP 1 vs 2) -> caught by a transfer check against existing artifacts, now automated; premature 'gate passed' claim -> refuted by expanded paired evaluation (F5). Each failure produced a reusable guard now built into the notebooks.

Deliverables

Final report (74 pp, appendices C-D) - Concepts & Corrections - Project Atlas - Vector Similarity Handbook - defense deck + cheat sheets - 25 runnable Colab notebooks (A1-F5) including V1, a reusable structure/semantics diagnostic battery for any $[n, d]$ representation - deployed artifact: 0.79 MB fp16 adapter, shared Qdrant index, 39 ms end-to-end demo path.

Open items

E1.2 isotropy-rescue run (embeddings ready, minutes); E1 scale sweep (dinov2-small/base/large with capacity controls) - would turn the two-point PRH sample into a slope; F3/F4 if moving the ceiling is ever worth trading the seconds-refit property; E3 internal-splice scaffold.